Reviewer Recommendation and Comments for Manuscript Number STEM-CELL-REPORTS-D-26-00194

Perinatal Nicotine Exposure Disrupts Hematopoietic Stem Cell Development and Elevates Influenza Susceptibility in Adulthood

Original Submission
Willem Buys

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Recommendation: Appropriate with Minor Revision

Transfer Authorization	**Response**
I consent to include my identifying information if the paper is transferred to another journal published by Cell Press or Elsevier.	Yes
In addition to comments to authors, I consent to the transfer of my comments to editors if the paper is transferred to another journal published by Cell Press or Elsevier.	Yes

Custom Review Question(s):	**Response**
Did anyone assist you with this review? If so, please give their name and position.	No

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Reviewer Comments to Author

Reviewer Summary: In this concise hypothesis-generating manuscript, Forsberg et al. report various changes of immune cell compositions mouse pups after pan-gestational enteral nicotine exposure of the dam, which are associated with an increase disease severity of subsequent influence infections. These results are interesting and the data quality is adequate. Data interpretation is mostly appropriate. Due to a lack of mechanistic data and a number of key mechanistic experiments, the novelty of these findings is only moderate, despite the study's strong rationale. I encourage the authors to continue this line of study to provide any mechanistic insight.

Recommendation: Publish with minor revisions.
Major Comments:
1. None.
Minor Comments:
2. Quantification and statistical analysis: What kind of outlier analysis was used and using which parameters? If outliers were removed, I highly recommend showing un-altered figures comprising all data points in the supplement, with outliers marked. Figure legends should also contain information on how many data points were removed as outliers.
3. "Perinatal" may be somewhat misleading, as nicotine was administered throughout pregnancy. The authors mention the layered nature of fetal hematopoiesis. This is of special importance to the development of tissue resident immune cell pools. It is thus especially interesting, whether the effects on lung macrophages (which are at least partly yolk sack derived) and bone-marrow HSC (dorsal aorta / AGM derived) are coupled or not. Due to their development during distinct embryonic and fetal stages, it may be possible to resolve this using different times of exposure.
4. Figure 1A, 2A, 5A, 6A: This figure suggests that dams continued to receive nicotine water after birth. This is not clearly distinguished in the respective methods section: Please clarify. I strongly recommend also discussing the potential of behavioral and other physiological changes of nicotine-exposed dams undergoing withdrawal during their off-spring's first days of life, which may cause unique effects on pups that are independent of nicotine exposure itself.
5. Fig. 1B: Visually, the control liver HSCs seem to cluster in two sub-populations: A low and a high-HSC number subgroup. The lower subgroup is on the same level as the nicotine exposed group. The higher subgroup is thus responsible for the observed significant difference of the mean. Please thoroughly analyze any documented characteristics of the higher vs lower groups to ensure that this is a random effect and not due to some unaccounted confounder. Alternatively, this may suggest a log-normal instead of a normal distribution. Please consider consequent use of non-parametric tests throughout the study.
6. In the absence of transplantation experiments, please use the term "central trained immunity" with special care.
7. Mothers absorbed nicotine enterally in this study, instead of the much more common pulmonary delivery in humans. Please discuss this as a potential limitation, considering systemic uptake, biodistribution, first pass, etc. Both nicotinergic and non-nicotinergic effects may also influence oxygen-delivery to the fetus in the real-world case, which is not being considered here.
8. Please discuss the expanded pool of lineage-negative non-HSPC you describe: What cells are these?
9. Page 8, "We collected serum (extracellular fluid) from the liver": Serum and extracellular fluid are not synonymous. Did you collect serum or extracellular fluid or a general mixed supernatant after isolating liver cells?
10. Page 13, "may be explained by their sequestration to the lung": You measured cell proportions in the lungs in this manuscript. Did you find increased numbers of NK-cells or is this a hypothesis based on the literature?
11. Visualization: I appreciate that the authors consequently overlaid single points. Displaying results as box plots with single point overlay may be even more appropriate than "dynamite plunger" plots with mean +/- SEM.
Language & Proofing:
12. Language use is excellent with only minor typos and spelling mistakes, which are likely accidental.

13. Continuous line numbers are appreciated during the review process.
14. Quantification and statistical analysis, line 5: There is a period after the +/- sign.
15. Page 12, line 4: Spelling.
16. Page 13, "independent of sex and of body weight": "~~of~~" -> "and".
17. Page 13, "infection and infiltrated into the respiratory": "~~infiltrated~~" -> "infiltrate"?
Further Suggestions & Future Directions:
18. Figure 3: While not statistically significant per cell type, there seems to be a trend towards lesser peripheral blood cell concentrations, which the authors may consider mentioning in the respective results subsection.
19. Consider supplementing this study with CFC assays of fetal liver and bone marrow-derived HSPC.
20. Analyzing elderly mice after in utero exposure and analyzing the first & second generation off-spring of mice exposed in utero would be highly interesting to capture the scope and persistence of epigenetic changes introduced by perinatal nicotine exposure. In addition, exposing dams before but not during pregnancy or only after birth but while breast feeding may be interesting. Adding transplantation experiments would further strengthen this line of inquiry.
21. Pages 8, 9: Consider analyzing which cells in the liver were responsible for the altered niche signaling, i.e., which cells produced these cytokines, e.g., by intracellular flow cytometry or, as a proxy, by transcriptomics.
22. Consider sequencing the bone marrow of in utero exposed off-spring, searching for typical, clonal hematopoiesis-associated mutations, which may explain some of the effects observed in this study.
23. Please add human data. This is a highly artificial model system.

Reviewer Confidential Comments to Editor:

Please rate the paper on a scale of 1 (low) to 10 (high): 5

Please indicate a likely rating for the paper if your comments were addressed: 8

As a guide, we would interpret a rating of 6 as an indication that the paper is of borderline suitability for Stem Cell Reports.

If you feel the paper would be more appropriate for another journal, please specify which: Possibly 'Blood'. Stem Cell Reports is only a decent fit in the absence of transplant experiments.

Did you find the organization of the data in the main paper and in the supplemental information to be appropriate? Yes.

Any other comments to the editor: This paper represents an excellent execution of a study with a strong rationale but a very limited scope. Without mechanistic data and without any human data, this is basically just a report of interesting findings in an artificial model system. With mechanistic data, this could become a paper of public interest beyond just the scientific sphere.

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